

Yu-Ting Wu

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https://kevincosner.github.io/

Research Interests

- Computer graphics, computer vision, VR/AR/MR, image processing, artificial intelligence

Education

- **National Taiwan University** Taipei, Taiwan
Ph.D. in Computer Science Sep. 2009 - June 2014
Advisor: Yung-Yu Chuang
Dissertation: Sampling and Reconstruction Techniques for Efficient Monte Carlo Rendering
- **National Chiao Tung University** Hsinchu, Taiwan
Master in Computer Science Sep. 2007 - June 2009
Advisor: Zen-Chung Shih
Thesis: Visibility-Guided Importance Sampling
- **National Chiao Tung University** Hsinchu, Taiwan
Bachelor in Computer Science Sep. 2003 - June 2007
Rank 1st in class, 6 times Academic Excellence Award (top 5%)
Honorary Member of Phi-Tau-Phi Scholastic Honor Society in 2007

Academic Experience

- **Associate Professor - National Taipei University**, New Taipei City, Taiwan Aug. 2026 - Present
Department: Computer Science and Information Engineering
- **Assistant Professor - National Taipei University**, New Taipei City, Taiwan Feb. 2022 - Jul. 2026
Department: Computer Science and Information Engineering
- **Postdoctoral researcher - National Taiwan University**, Taipei, Taiwan Feb. 2020 - Jul. 2021
Host: Yung-Yu Chuang

Industry Experience

- **Technical Manager - MediaTek Inc.**, Taipei, Taiwan Sep. 2021 - Jan. 2022
- **Senior Algorithm Developer - Toppano Inc. (startup)**, Taipei, Taiwan May 2018 - Jan. 2020
- **Principal Engineer - HTC Inc.**, New Taipei City, Taiwan Sep. 2014 - Apr. 2018

Publications (*: the corresponding author)

- **Orthographer: Generating Orthographic-Style Projections for Elongated Architectural Structures**
Yu-Hsuan Hsieh, **Yu-Ting Wu***
In Computer Graphics Forum, to appear.
[SCIE, JCR 2024 IF: 2.900, Computer Science, Software Engineering, Ranking 37.98%]
- **Enhanced 360-Degree Panorama Synthesis: Leveraging Smartphone Sensors and Semantic-Driven Neural Architecture**
Yu-Wen Liu, Nien-Hsin Tseng, Hsin-Chang Yu, **Yu-Ting Wu***
In Journal of Information Science and Engineering, to appear.
[SCIE, JCR 2024 IF: 1.100, Computer Science, Software Engineering, Ranking 77.13%]
- **Preserving Photographic Defocus in Stylised Image Synthesis**
Hong-Yi Wang, **Yu-Ting Wu***
In Computer Graphics Forum, to appear.
[SCIE, JCR 2024 IF: 2.900, Computer Science, Software Engineering, Ranking 37.98%]

- **Efficient Stereo-Aware Screen-Space Ambient Occlusion with Adaptive Computation**
Yu-Ting Wu*
In IEEE Computer Graphics and Applications, to appear.
[SCIE, JCR 2024 IF: 1.400, Computer Science, Software Engineering, Ranking 64.34%]
- **Efficient Environment Map Rendering Based on Decomposition**
Yu-Ting Wu*
In Computer Graphics Forum, volume 44, issue 1, February 2025.
[SCIE, JCR 2024 IF: 2.900, Computer Science, Software Engineering, Ranking 37.98%]
- **StylePart: Image-based Shape Part Manipulation**
I-Chao Shen, Li-Wen Su, Yu-Ting Wu, Bing-Yu Chen
In The Visual Computer, volume 44, pages 67–78, January 2025.
[SCIE, JCR 2024 IF: 2.900, Computer Science, Software Engineering, Ranking 37.98%]
- **Improving Cache Placement for Efficient Cache-based Rendering**
Yu-Ting Wu*, I-Chao Shen
In The Visual Computer, volume 40, pages 8173–8187, November 2024.
[SCIE, JCR 2024 IF: 2.900, Computer Science, Software Engineering, Ranking 37.98%]
- **360MVSNet: Deep Multi-view Stereo Network with 360° Images for Indoor Scene Reconstruction**
Ching-Ya Chiu, Yu-Ting Wu, I-Chao Shen, Yung-Yu Chuang
In Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2023, Waikoloa, Hawaii, January 2023.
- **StyleFaceUV: a 3D Face UV Map Generator for View-Consistent Face Image Synthesis**
Wei-Chieh Chung*, Jian-Kai Zhu*, I-Chao Shen, Yu-Ting Wu, Yung-Yu Chuang (*: joint first authors)
In Proceedings of British Machine Vision Conference (BMVC) 2022, London, UK, November 2022.
- **ScannerNet: A Deep Network for Scanner-Quality Document Images under Complex Illumination**
Chih-Jou Hsu, Yu-Ting Wu, Ming-Sui Lee, Yung-Yu Chuang
In Proceedings of British Machine Vision Conference (BMVC) 2022, London, UK, November 2022.
- **Learning to Cluster for Rendering with Many Lights**
Yu-Chen Wang, Yu-Ting Wu*, Tzu-Mao Li, Yung-Yu Chuang
In ACM Transactions on Graphics (Proceedings of ACM SIGGRAPH Asia 2021), volume 40, number 6, article 277, Tokyo, Japan, December 2021.
[SCIE, JCR 2021 IF: 7.403, Computer Science, Software Engineering, Ranking 8.18%]
- **Multi-Resolution Shared Representative Filtering for Real-Time Depth Completion**
Yu-Ting Wu, Tzu-Mao Li, I-Chao Shen, Hong-Shiang Lin, Yung-Yu Chuang
In Proceedings of High-Performance Graphics (HPG) 2021, Online, July 2021.
- **ClipFlip: Multi-view Clipart Design**
I-Chao Shen, Kuan-Hung Liu, Li-Wen Su, Yu-Ting Wu, Bing-Yu Chen
In Computer Graphics Forum, volume 40, number 1, pages 327-340, February 2021.
[SCIE, JCR 2021 IF: 2.363, Computer Science, Software Engineering, Ranking 48.18%]
- **Dual-Matrix Sampling for Scalable Translucent Material Rendering**
Yu-Ting Wu, Tzu-Mao Li, Yu-Hsun Lin, Yung-Yu Chuang
In IEEE Transactions on Visualization and Computer Graphics (TVCG), volume 21, number 3, pages 363-374, March 2015.
[SCIE, JCR 2015 IF: 1.400, Computer Science, Software Engineering, Ranking 23.58%]
- **VisibilityCluster: Average Directional Visibility for Many-Light Rendering**
Yu-Ting Wu, Yung-Yu Chuang
In IEEE Transactions on Visualization and Computer Graphics (TVCG), volume 19, number 9, pages 1566-1578, September 2013.
[SCIE, JCR 2013 IF: 1.919, Computer Science, Software Engineering, Ranking 12.38%]
- **SURE-based Optimization for Adaptive Sampling and Reconstruction**
Tzu-Mao Li, Yu-Ting Wu, Yung-Yu Chuang

In ACM Transactions on Graphics (Proceedings of ACM SIGGRAPH Asia 2012), volume 31, number 6, article 194, Singapore, November 2012. *(selected as a highlight paper by the chair)*.
[SCIE, JCR 2012 IF: 3.361, Computer Science, Software Engineering, Ranking 0.95%]

Preprints, Workshop Papers, Short Papers, Posters

- **VisibilityChunk: Average Directional Visibility for Importance Sampling**
Yu-Ting Wu, Yung-Yu Chuang
In ACM SIGGRAPH Asia 2012 Poster, article 44, Singapore, November 2012 *(selected as a highlight poster by the chair)*.
- **Improved Reflective Shadow Maps with Visibility Approximation**
Mifan Bang, **Yu-Ting Wu**, Yung-Yu Chuang
In Computer Graphics Workshop (CGW 2011), Taipei, Taiwan, July 2011.
- **Horizon Occlusion Culling for 3D Navigation**
Yun-Feng Chou, **Yu-Ting Wu**, Shiang-Chun Chang, Mu-Heng Li, I-Chen Lin, Zen-Chung Shih, Rung-Ren Lin
In International Workshop on Advanced Image Technology 2008 Poster, Hsinchu, Taiwan, January 2008.

Patents

- **Electronic device, method for displaying an augmented reality scene and non-transitory computer-readable medium**
Yu-Ting Wu, Ching-Yang Chen
ROC Patent No: I711966, December 01, 2020
US Patent No: 10636200, April 28, 2020
- **Virtual reality device, image processing method, and non-transitory computer-readable medium**
Yu-Ting Wu, Chun-Wen Cheng, Ching-Yang Chen
ROC Patent No: I684163, February 01, 2020
- **Three-dimensional modeling method and electronic apparatus thereof**
Sheng-Jie Luo, Liang-Kang Huang, **Yu-Ting Wu**, Tung-Peng Wu
US Patent No: 10152827, December 11, 2018

Grants

- **Illumination Reconstruction Algorithms for Augmented and Mixed Reality (3/3)**, NSTC, PI, Aug. 2026 - Jul. 2027
- **Illumination Reconstruction Algorithms for Augmented and Mixed Reality (2/3)**, NSTC, PI, Aug. 2025 - Jul. 2026
- **Illumination Reconstruction Algorithms for Augmented and Mixed Reality (1/3)**, NSTC, PI, Aug. 2024 - Jul. 2025
- **Photo-realistic Rendering Algorithms for Extended Reality and Mobile Devices (2/2)**, NSTC, PI, Aug. 2023 - Jul. 2024
- **Photo-realistic Rendering Algorithms for Extended Reality and Mobile Devices (1/2)**, NSTC, PI, Aug. 2022 - Jul. 2023

Honors and Awards

- Honorable Mention in Undergraduate Project Contest 2026 (as an advisor), National Taipei University, 2026
- Outstanding Teaching Award (accumulated three Teaching Excellence Awards), National Taipei University, 2025
- Academic Research Performance Incentive 2025, National Taipei University, Dec. 2025

- 2nd place in Undergraduate Project Contest 2025 (as an advisor), National Taipei University, 2025
- Teaching Excellence Award, National Taipei University (top 4%), 2024
- Teaching Excellence Award, National Taipei University (top 4%), 2023
- Annual Excellent Advisor, National Taipei University, 2023
- 3rd place in Undergraduate Project Contest 2023 (as an advisor), National Taipei University, 2023
- Highlight Paper, SIGGRAPH Asia 2012
- Highlight Poster, SIGGRAPH Asia 2012 Poster
- Honorary Member, Phi-Tau-Phi Scholastic Honor Society, 2007
- Academic Excellence Award, National Chiao Tung University, Fall 2007
- Master Freshman Scholarship, National Chiao Tung University, 2007
- 3rd place in Communication Competition Contest, Ministry of Education, 2006
- 2nd place in Computer Science Project Competition, National Chiao Tung University, 2006
- Academic Excellence Award, National Chiao Tung University, Fall 2006
- Academic Excellence Award, National Chiao Tung University, Spring 2006
- Academic Excellence Award, National Chiao Tung University, Fall 2005
- Academic Excellence Award, National Chiao Tung University, Spring 2005
- Academic Excellence Award, National Chiao Tung University, Fall 2004
- Academic Excellence Award, National Chiao Tung University, Spring 2004
- Academic Excellence Award, National Chiao Tung University, Fall 2003

Professional Services

- **Reviewer**
 - PG 2025, AAAI 2025, SIGGRAPH 2024, CVPR 2024, CVPR 2023, ACCV 2022, ICCV 2021, APMAR 2021, CVPR 2020, ECCV 2020, WSCG 2013, IEEE TMM, CGF, TVC, IJPRAI, JISE
- **Invited Talk**
 - **Virtual Reality: Technology and Content Development,** *May 2016*
National Cheng Kung University, Tainan, Taiwan
Yuan Ze University, Taoyuan, Taiwan
 - **Introduction to Physically-Based Ray Tracing,** *Dec. 2013*
Industrial Technology Research Institute (ITRI), Hsinchu, Taiwan